LUMYVUE V6 - finalized model and application overview

Written so that another AI assistant (or a reviewer) can understand what the system is, how the final model works, how it was evaluated, and what every screen of the Windows app does. Status as of 2026-11-06, branch v6-integration. Photographs shown in the app are never reproduced here; no screenshots are included (see section 9).

1. What the system is

LUMYVUE (project folder SkinDevApp; the hardware simulator lives in a separate repository, PrecisionSkinSimulator) is a research prototype of an AI-assisted facial-skin screening kiosk. A Windows WPF application (.NET Framework 4.7.2) runs on a touchscreen mini PC with a webcam and a QR701 58 mm thermal receipt printer. A participant is registered, gives consent, and is photographed from the front, left and right. The AI analyses each photograph and the app shows one result per photograph, evidence maps, a printed receipt, an A4 research report and review screens for a licensed dermatologist.

It is a screening aid, not a diagnosis. It is not clinically validated, and every assessment waits for dermatologist review.

Conditions: three detectable skin conditions - Acne, Hyperpigmentation and Eczema (multi-label: more than one can be present). 'Normal' is never a model output; it is derived: no finding is supported, every detector worked, no flag was raised and the photo quality was assessed.

2. The finalized model (V6, run open_run02)

V6 gives ONE unified result per photograph. It replaced showing two separate answers (a whole-photo classifier and separate lesion detectors) that could disagree. The previous system, V5 (a single MobileNetV2 classifier, Acne / Hyperpigmentation / Eczema / Normal), is kept inside the app as the baseline and for rollback but is not shown on screen when V6 is on.

2.1 Pipeline for one photograph

Step	What happens	Output
1. Whole-photo CNN	MobileNetV2 on the whole photo. Preprocessing: BGR to RGB, bilinear resize to 224x224, (pixel / 127.5) - 1.	128-number embedding + 3 scores (one per condition)
2. Three YOLO detectors	One specialist detector per condition (YOLOv8, ONNX), each an ensemble of up to 3 models whose boxes are merged by weighted box fusion. A candidate pool is kept down to confidence 0.05, with NO face filter (training evidence was built without it).	Boxes per condition; 21 evidence numbers (7 per condition: detector available, detected, best deployed confidence, best pool confidence, sqrt of relative area, centre x, centre y)
3. Region branch (full variant only)	Up to 12 regions chosen from the detector boxes (tier 1 deployed boxes round-robin across conditions, tier 2 lower-confidence candidates, tier 3 wider-context copies). Each is cropped square with 25% margin (60% for tier 3), resized to 224, and read by a second MobileNetV2.	Per region 83 numbers: 64 embedding + 4 scores (Acne, Hyperpigmentation, Eczema, Background) + 15 geometry/evidence values; a mask marks real regions
4. Fusion network	A trained network takes the Global embedding and logits, the 21 evidence numbers and (full variant) the region features with masked attention pooling. Nothing is averaged or voted by hand.	3 raw fusion scores
5. Calibration	Platt scaling per condition: p = sigmoid(a*z + b).	3 calibrated probabilities
6. Thresholds	Present if p >= threshold + band; Not found if p < threshold - band; otherwise Uncertain. Band = 0.0. global_yolo thresholds 0.42 / 0.37 / 0.27; full thresholds 0.43 / 0.34 / 0.33 (Acne / Hyperpigmentation / Eczema).	Present / Not found / Uncertain per condition

Step	What happens	Output
7. Side checks and flags	A second fusion network with the detector evidence hidden raises a CNN-vs-detector conflict flag (it never changes a score). Other flags: near threshold, out-of-distribution (Mahalanobis distance of the embedding), photo-quality problems.	Review flags
8. Detector-mark rule	If the fusion says Not found but the detector marked an area for that condition, the verdict is 'Possible' and the overall result cannot be 'no findings'. The detector never overrides a Present.	Possible

2.2 Two deployed variants (App.config key V6Variant)

- **global_yolo (default):** steps 1, 2, 4-8 (Global CNN + YOLO evidence into the fusion network). Faster.
- **full (optional):** adds the region branch (step 3). About 2 s per photograph on CPU in tests. It showed no statistically clear gain over global_yolo, so global_yolo was chosen as the default. If the full models cannot load, the app falls back to global_yolo; every saved assessment records which variant ran.

2.3 Overall result and states

One headline per photograph: FINDINGS_SUPPORTED, MULTIPLE_FINDINGS_SUPPORTED, NO_FINDINGS_SUPPORTED, UNCERTAIN_REVIEW_REQUIRED, RECAPTURE_REQUIRED (blocking photo-quality problem), OUT_OF_SCOPE_REVIEW_REQUIRED (unlike the training photos) or INFERENCE_ERROR (never a normal-looking result). Any doubt - an Uncertain condition, a flag, a missing detector - means review. Absence ('Not found') counts only when that condition's detector was available and did not mark anything.

2.4 Evidence maps

Heat maps are detector-evidence maps: a Gaussian bump per detector box (strength = confidence / 0.70, capped at 1), drawn only inside the face region, in the condition's colour (Acne red, Hyperpigmentation blue, Eczema yellow). They show where the detectors found evidence; they are NOT an explanation of the classifier (a Grad-CAM through the fusion model was prototyped and rejected because the maps were 95-98% identical across conditions). Each scan saves the original, the heat map, the marked areas, a combined view and a sealed JSON (v6_assessment.json, SHA-256 seal, schema lumyvue-v6-assessment/1.0).

3. Files that make up the model

File (SkinDevApp/Model/V6 unless noted)	Role
v6_global_features_mobilenetv2.onnx	Whole-photo MobileNetV2: embedding (128) + logits (3)
v6_fusion_global_yolo_seed42.onnx	Fusion network used by the default variant
v6_fusion_full_seed42.onnx + v6_roi_features_mobilenetv2.onnx	Full variant: region MobileNetV2 and its fusion network
v6_fusion_global_only_seed42.onnx	Used only for the CNN-vs-detector conflict flag
v6_ood_model.json	Mean and precision matrix for the out-of-distribution check
v6_deployment_config.json	Model hashes, calibration (Platt a, b), thresholds, band, detector thresholds, ROI constants
Model/lesion_detector_<condition>.onnx (+ _m2, _m3) and .json	The three YOLO detector ensembles (shared with V5)
Model/precisionskin.onnx	V5 classifier, kept as baseline and rollback

Every V6 model file is verified against its recorded SHA-256 at load; a mismatch makes V6 refuse to run. The feature flag V6Unified (App.config; default on) switches the app between V6 and plain V5.

4. Evaluation and honest status

Evaluated on a 'fair set' of 132 fully verified validation photographs that V5 had not seen: 47 hyperpigmentation-positive, 26 with no condition at all, 18 eczema-positive. The locked test split was never opened (it has no labels). Development data only: validation was used to choose models and stop training, so V6 may look slightly better here than on fresh photographs.

System	Hyper. missed (of 47)	Clean flagged hyper. (of 26)	F1 acne	F1 hyper.	F1 eczema	Macro-F1
V5 argmax (baseline)	9	1	0.835	0.658	0.718	0.737
V5, tuned thresholds	7	2	0.836	0.725	0.667	0.743
Global only	10	3	0.852	0.701	0.647	0.733
V6 Global + YOLO (default)	10	2	0.870	0.711	0.632	0.737
V6 full fusion (optional)	8	2	0.862	0.759	0.645	0.755
ROI only	9	1	0.830	0.734	0.629	0.731

- **Pre-registered criterion not met:** no clear improvement of V6 over V5. Every paired 95% bootstrap interval against V5 (macro-F1, hyperpigmentation missed rate, false-alarm rate) includes zero.
- V6 is not better at eczema at the calibrated cutoff (F1 0.63-0.65 vs V5 0.72), although it ranks eczema better (AP 0.75 vs 0.66, on only 18 positives). V5 with thresholds tuned on the fusion split already matches most of the gain.
- The region branch adds nothing measurable over Global + YOLO; the fusion abstains on about 79% of Normal photographs.
- Detectors alone: about 91% hyperpigmentation recall but about 27% of clean photographs flagged. This is why a detector mark with a fusion 'Not found' is shown as 'Possible'.
- **What V6 does deliver:** one unified, auditable result (sealed per-scan record, evidence maps, review flags, no contradicting second answer), with measured accuracy comparable to V5 on a small set. It is not shown to be more accurate.
- Not done: independent test data, dermatologist validation, consent for external data, real-camera observation of the full variant. Hyperpigmentation generalisation is the weak spot (it works on the user's tested photographs but no larger measured result exists).

4.1 Software verification

- C# vs the Python reference detector (on 13 real photographs; the reference reproduces the training-time detection cache exactly): 270 boxes agree to 0.0001 px.
- Fusion CPU parity: max difference 1.9e-6; 0 of 1200 decisions flipped. Full variant against its Python reference: calibrated probabilities differ by at most 1e-6, 0 of 39 decisions flipped.
- 97 automated tests pass (xunit, net10.0, linking the production V6 sources). The user's exe is only rebuilt on request.

5. Live camera behaviour

The camera screen shows a face guide (Move closer, Center your face, ...). A state machine drives Live -> Scanning -> Stabilizing -> Hold Still -> Stable-Capturing -> Final Analysis -> Results; the fast V5 loop still decides when the picture is steady for auto-capture. V6 live reading (background loop, newest frame wins, about 2-3 s per reading) updates the on-screen bars and verdict and draws a region-limited evidence overlay. Uploaded images (researcher) go through the same V6 assessment; a missing face is a warning for uploads and blocking for live capture.

6. Application screens (every form and window)

Navigation: MainWindow hosts a Frame. It starts on Login, then Home (dashboard). Everyone who signs in can use every screen (all accounts are treated as the Researcher role). Styling uses a warm olive/cream theme (Themes/Lumyvue.*.xaml). Touch-friendly: LUMYVUE's own on-screen keyboard and dialogs.

Screen (file)	What it shows / does
Splash (Views/SplashWindow.cs)	Start-up screen; real steps reported: Initializing system, Connecting camera, Preparing AI models, Loading workspace; then fades into the main window.
MainWindow (MainWindow.xaml)	Shell window with a navigation Frame, 'Closing LUMYVUE' overlay and page-transition animation.
Login (LoginForm.xaml)	Sign in with username and password (confirm-password when creating an account). Tag lines: 'AI skin health monitoring', 'Three angles, one clear picture of your skin.' and a not-a-diagnosis notice.
Home / Dashboard (Views/HomePage.xaml)	Workspace tiles: New registration; Skin analysis (choose a patient, take front/left/right photos); Patient records; Research dashboard; Waiting for review (pending count); Model information; Settings; About LUMYVUE; Old report log. Also a 'Check system again' readiness panel and the research-prototype disclaimer.
New registration (RegistrationPage.xaml)	Patient registration following the clinic's Patient's Chart: personal information, present illness, personal/past/family medical history, optional physical examination and special tests. Automatic PatientID. Signature is on a separate paper consent form. Edit mode (from Patient records) requires a reason; scan results are never touched.
Consent (ConsentPage.xaml)	Research and facial-image consent. Scanning stays blocked until the required items are explicitly ticked; each decision is stored with a timestamp.
Participant picker (Views/ParticipantPickerWindow.cs)	Skin analysis needs an existing participant: search by PatientID or name, pick the match, Continue. Searching never creates a record.
Preparation (PrepearationPage.xaml)	Camera check and patient reminders before scanning.
Skin analysis / live camera (ClientDashboardForm.xaml)	Live camera with face guide, centre lines option, auto-capture toggle, three-photograph mode (front, left, right), live AI analysis bars (Acne, Hyperpigmentation, Eczema, Normal) and a verdict, zoom, retake per view, 'Accept views and view results', pending reviews, researcher view, comparison gallery, upload-and-analyse (researcher), capture details/review, Retry save, Go to Consent.
Final results (Views/ResultsPage.xaml)	One saved scan session, read only from the database and files (never re-run): the 'Unified AI assessment' card (headline, one row per condition with Present / Possible / Not found / Uncertain, AI support vs decision limit, whether the detector marked an area, review notes, disclaimer) and 'The photograph' panel with view tabs (Front, Left, Right) and toggles Original / Heat map / Marked areas / Both; per-view summary; receipt and report actions.
Comparison gallery (ComparisonGalleryWindow.cs)	Researcher comparison for one scan: tabs Overall / Front / Left side / Right side, V6 evidence maps first (V5 classifier maps are still listed as 'Classifier maps (V5)').
Researcher result (ResearcherResultWindow.xaml)	Researcher view of one archived capture and its Pending Review state.
Researcher verification (Researcher VerificationWindow.xaml)	AI prediction next to two independent human layers: the researcher reference assessment and the licensed dermatologist validation. Both are append-only rows in their own tables; AI rows are never modified.
Dermatologist review (Views/DermatologistReviewWindow.cs)	Clinical summary for the licensed dermatologist without raw AI metrics: who/when, one plain AI-assisted summary, per-view table, the three views (original first, toggles for AI region of interest), then the dermatologist's own assessment. Never pre-filled, never a diagnosis.
Patient records (PatientRecordPage.xaml)	Participant Records (researcher/admin): search by PatientID; information, skin profile, consent, latest result and full analysis history; selecting a scan shows its exact saved data. Has a 'waiting for review' filter.
Research dashboard (ResearchDashboardPage.xaml)	Summaries counted only from stored data: single-hue horizontal bars with counts and shares; Refresh button.
Research report (Views/ResearchReportWindow.cs)	'PrecisionSkin Research Analysis Report' (A4) for one saved scan: unified assessment summary, per-view detector evidence pages, built only from stored data.

Screen (file)	What it shows / does
Receipt preview (Views/ReceiptPreviewWindow.cs)	The thermal receipt as it will print on the QR701 (57.5 mm paper, 48 mm / 384-dot print width) with Print / Cancel / Close. Printing needs a deliberate press; a repeat within a minute asks first. 'Sent to printer' is never shown as 'printed'.
Receipt history (Views/ReceiptHistoryWindow.cs)	Every receipt version of one scan and every print attempt, newest first; earlier versions can be reprinted by a researcher.
Settings (SettingsPage.xaml)	Tabs for camera and capture preferences, the thermal receipt printer, accounts and the change history. Save settings / Restore defaults; every changed value is audited. Image-analysis, heat-map and lesion-marking values are fixed for the study.
Model information (ModelInfoPage.xaml)	Traceable model identity (researcher/admin): which model files are in use, readiness.
About (Views/AboutWindow.cs)	What the system does, how to read its output and its limits.
Old report log (UserDashboardForm.xaml)	Legacy, read-only log of reports saved by earlier versions (PatientLogs): total evaluations, unique patients, ONNX engine status. New scans are in Patient records.
Legacy print report (PrintReportWindow.xaml)	Older single-page PDF report layout (patient name, consultation date, class percentages, recommended next steps, disclaimer). Superseded by the research report and receipt.
Dialogs and keyboard (Views/LumyDialog.cs, Views/OnScreenKeyboard.cs)	LUMYVUE's own message dialog (replaces MessageBox) and an on-screen keyboard that slides up when a text box is tapped (the mini PC has no physical keyboard).

7. What is saved for each scan

- Original photographs (front, left, right) and the saved V5 scores for historical scans.
- V6: v6_assessment.json (sealed with SHA-256; includes the candidate pool, flags, variant and model hashes), v6_overlay_<Condition>.png, v6_heatmap_<Condition>.png, v6_overlay_all.png, v6_combined_<Condition>.png, v6_marked_areas.png.
- Receipt versions and print attempts, researcher and dermatologist assessments (append-only), consent timestamps, settings change history.
- Historical scans always show the output of the model that produced them; nothing is recomputed when a record is opened.

8. Safety and honesty rules built into the app

- Not a diagnosis; every assessment shows a research disclaimer and waits for dermatologist review.
- 'Not found' does not rule out a skin condition. Boxes are possible locations only and are not a lesion count or severity.
- Missing detector or failed inference is never shown as normal (INFERENCE_ERROR / uncertain).
- Labels are never falsified; the locked test split is never used; AI rows are never edited by reviewers.

9. Notes for the reader

- Screenshots were not included in this PDF; each screen above is described from its source file. Screenshots can be added from the running app.
- Technology: C# / WPF (.NET Framework 4.7.2), OpenCvSharp, Microsoft.ML.OnnxRuntime. Training: Python (TensorFlow 2.x / Keras, YOLOv8), exported to ONNX; training notebooks live in v6/ (open_run02 is the final run).
- Defense date 2026-11-16. Remaining work is not model training but validation: a larger fair set, a hyperpigmentation detector re-measurement on hand-boxed photographs, independent testing and dermatologist review.